

SECTION C**11 (long)**

(a) The response $y(t)$ of a linear system with input $f(t)$ is given by the convolution integral

$$y(t) = \int_0^t g(t - \tau)f(\tau)d\tau$$

where $g(t)$ is the impulse response. Both $f(t)$ and $g(t)$ are zero for $t < 0$. Show that

$$\int_0^t g(t - \tau)f(\tau)d\tau = \int_0^t f(t - \tau)g(\tau)d\tau \quad [3]$$

(b) The response of the linear system is governed by the differential equation

$$\frac{d^2y}{dt^2} + 14\frac{dy}{dt} + 49y = f(t)$$

(i) Compute the step response. [10]

(ii) Show that the impulse response is given by

$$g(t) = te^{-7t} \quad [2]$$

(iii) Find the response to an input

$$f(t) = \begin{cases} 0 & t < 0 \\ 1 & 0 \leq t \leq 1 \\ 0 & t > 1 \end{cases}$$

Sketch the response and mark its salient features. [15]